

SYSTEM REQUIREMENTS SPECIFICATION (CAHIER DES CHARGES)

AI Risk Prediction & Early Triage Assistant

TetraTHON 2026 · HT1 HealthTech Track · Team 360 HORIZON (TETRA004)

TEAM ID	TEAM NAME	INSTITUTIONAL ALLIANCE	REPOSITORY
TETRA004	360 HORIZON	NUV · ISEN Méditerranée	github.com/boudjellal-malak/TETRA004

1. CONTEXT & PROBLEM STATEMENT

Lifestyle-related chronic conditions—specifically Diabetes, Coronary Artery Disease (CAD), Chronic Kidney Disease (CKD), and Stroke—are experiencing a severe surge in global incidence, presenting an overwhelming burden on resource-constrained healthcare environments. The primary clinical bottleneck lies in delayed diagnosis: the vast majority of patients are identified only after irreversible secondary complications have already developed.

This structural delay is driven by concrete operational challenges in primary care settings, including a lack of clinical decision-support systems (CDSS), restricted consultation time windows, and fragmented access to updated evidence-based triage guidelines.

2. PROJECT OBJECTIVE

To engineer an Artificial Intelligence decision-support platform that empowers healthcare professionals to identify high-risk patients early using routine, non-invasive clinical indicators, while providing actionable, explainable triage and referral recommendations.

3. FUNCTIONAL SCOPE

3.1 In-Scope Features

- **Clinical Data Input Interface:** Intuitive entry form for patient baseline parameters (demographics, vital signs, clinical history, and optional routine laboratory test results).
- **Multi-Pathology Risk Scoring:** Independent risk computation across four target pathologies: Diabetes, Coronary Artery Disease, Stroke, and Chronic Kidney Disease.
- **Consolidated Clinical Dashboard:** Real-time stratification displaying categorized risk levels (**Low** / **Moderate** / **High** / **Urgent**) per condition.
- **Explainable AI (XAI) Engine:** Patient-level prediction breakdown identifying key risk drivers using SHAP (SHapley Additive exPlanations).
- **Missing Test Recommendations:** Intelligent suggestion of missing diagnostic tests required to refine prediction accuracy.
- **Urgent Triage Visual Alerts:** Automated visual notification system triggering high-priority red alerts for critical risk thresholds requiring immediate referral.

3.2 Out-of-Scope (Current Sprint Version)

- User authentication, role-based access control, and user account management.
- Persistent database storage of historical Electronic Health Records (EHR).
- Direct integration with live Hospital Information Systems (HIS / EHR middleware).

- Definitive automated medical diagnosis—the platform functions strictly as a screening and clinical decision-support tool, preserving practitioner clinical autonomy.

4. DATASETS & DATA PIPELINE

TARGET PATHOLOGY	DATASET NAME	SOURCE / PROVENANCE	SAMPLE SIZE (RECORDS)
Diabetes	Pima Indians Diabetes Dataset	Public Medical Repository	768
Coronary Artery Disease	Framingham Heart Study	Framingham Cohort Study	4,240
Stroke	Stroke Prediction Dataset	Public Medical Repository	5,110
Chronic Kidney Disease	UCI CKD Dataset	Apollo Hospitals, India	400

KNOWN DATA LIMITATIONS & MITIGATIONS

While these public datasets are benchmarks in academic research, they contain known demographic biases and legacy sampling limitations (e.g., Framingham cohort demographics). Accessing fully accredited clinical data repositories (such as PhysioNet or MIMIC-IV) requires formal institutional credentialing that exceeded the 36-hour hackathon sprint timeline. Acquiring accredited data pipelines is explicitly designated as a core post-hackathon roadmap objective.

5. TECHNICAL CHOICES & MACHINE LEARNING ARCHITECTURE

5.1 Model Architecture: XGBoost Engine

XGBoost (eXtreme Gradient Boosting) was selected as the primary algorithmic engine due to its superior performance on tabular clinical data, native missing value handling, and seamless integration with tree-based SHAP explainability models. Rather than training a single multi-label model, four independent specialized models were constructed to prevent signal dilution caused by heterogeneous clinical feature spaces.

5.2 Native Handling of Missing Data

Physiologically impossible zero values (e.g., systolic blood pressure = 0) were explicitly converted to true missing values (NaN), allowing XGBoost to learn optimal default directional splits without introducing imputation bias.

5.3 Class Imbalance Strategy

To compensate for heavy class imbalance inherent in real-world epidemiology (e.g., 4.9% positive stroke incidence), models were parameterized using dynamic `scale_pos_weight` adjustments to increase recall sensitivity for critical positive cases.

5.4 Model Explainability (SHAP)

Every prediction is accompanied by a localized SHAP (TreeExplainer) attribution graph, mapping feature contributions (positive or negative) to the individual patient’s risk score. This transparency is vital for establishing clinical trust.

5.5 Disease-Specific Decision Thresholds

Decision boundaries are customized based on clinical risk asymmetry rather than a default 50% cutoff. For instance, Stroke detection employs a lower classification threshold (0.15) to maximize clinical recall (minimizing false negatives), whereas Diabetes and CKD maintain standard 0.5 thresholds.

6. SYSTEM ARCHITECTURE & TECH STACK

Component	Technology Stack	System Role & Responsibility
Frontend	React (Vite), Tailwind CSS	Dynamic clinical entry forms, risk display dashboard, SHAP feature visualization
Backend API	FastAPI (Python 3.10+)	RESTful prediction endpoints, input validation, ML pipeline orchestration
ML Models	XGBoost, Scikit-Learn	5 trained classifiers (4 disease models + 1 early CKD screening variant)
Explainability	SHAP (TreeExplainer)	Real-time compute of feature attribution scores per prediction API call

7. EXPERIMENTAL RESULTS & PERFORMANCE METRICS

Pathology	Recall / Sensitivity	ROC-AUC / PR-AUC	Clinical Benchmark Evaluation
Diabetes	68.5%	0.83 (ROC-AUC)	Robust, stable baseline performance
Coronary Artery Disease	Threshold Dependent	0.70 – 0.73 (ROC-AUC)	Consistent with published Framingham literature
Stroke	82.0% (Adjusted Threshold)	0.28 (PR-AUC)	Optimized for high-recall early screening
CKD (Early Screening Variant)	96.0%	99.27% Accuracy	High-accuracy detection using basic parameters only

8. LIMITATIONS & FUTURE ROADMAP

- **Clinical Data Migration:** Secure credentialed access to PhysioNet and MIMIC-IV datasets in collaboration with partner healthcare institutions.
- **Enterprise Security & EHR Storage:** Implement OAuth2 authentication, HIPAA-compliant encryption, and persistent patient record storage.
- **Clinical Validation Study:** Conduct formal clinical trial validation and usability reviews with medical professionals prior to deployment.
- **Modular Expansion:** Scale the architecture to support additional cardiometabolic and respiratory conditions.

9. PROJECT DELIVERABLES

- **Functional Prototype:** Fully integrated frontend and backend system demonstrated live during the evaluation round.
- **Public Source Code:** Hosted at github.com/boudjella1-malak/TETRA004.
- **Presentation Deck:** Official pitch deck covering problem space, technical approach, and business viability.
- **Technical Specification Document:** This formal system design document (Cahier des Charges).